

Beyond the Nearest Actor: A Scenario-Disjoint Audit of Current-Frame ODD-Context Proxies in WOMD

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Abstract

001 Current-frame TTC screens often compress a multi-
002 agent scene into the SDC and one focal actor, poten-
003 tially omitting observable roadway and traffic context.
004 We conduct a construct-controlled, scenario-disjoint au-
005 dit of whether dataset-observable ODD-context proxies
006 provide conditional validity beyond a nearest-clearance
007 SDC-actor baseline. On a processed WOMD v1.3.1
008 cohort of 18,445 scenarios (1.67M frames), we pre-
009 lock all development choices and evaluate frozen mod-
010 els once on an internal holdout of 2,804 scenarios
011 (255,164 frames). The binary reference marks frames
012 whose minimum linear-extrapolated swept OBB-TTC
013 between the SDC and any eligible dynamic actor
014 within 70 m is at most 3 s. With scenario-equal
015 weighting and paired scenario-block bootstrap infer-
016 ence, the full context model improves holdout AP
017 from 0.3224 to 0.3370 ($\Delta\text{AP} = +0.0147$; 95% CI
018 $[+0.0005, +0.0285]$). Family-restricted comparisons lo-
019 calize this increment to non-focal, SDC-relative in-
020 teraction summaries ($\Delta\text{AP} = +0.0161$; 95% CI
021 $[+0.0057, +0.0264]$); adding static-roadway or traffic-
022 composition features alone does not improve AP. As a
023 temporal completeness check, the pre-specified closing-
024 pressure-sum anchor remains positively associated with
025 target Peak ($\rho = +0.1914$), AUC ($\rho = +0.2345$), and
026 TET3 ($\rho = +0.0728$) after development-fitted focal-
027 kinematic adjustment, although magnitudes and sup-
028 port vary across features. Because the interaction prox-
029 ies and TTC target share current-state geometry, these
030 results support modest conditional and convergent sur-
031rogate validity within the processed cohort, not causal,
032 orthogonal, crash, or closed-loop safety validity. The
033 audit provides a bounded evaluation practice for test-
034 ing whether focal scene reductions omit measurable
035 current-frame context.

1. Introduction

Pairwise TTC surrogates characterize instantaneous criticality between two actors, while recorded driving scenes contain additional nearby actors and contextual conditions [3, 4, 8, 15]. Such a reduction is interpretable, but it may omit measurable information carried by other nearby actors, traffic composition, and roadway context [5, 6, 10]. We therefore ask a narrower evaluation question: after controlling for a clean nearest-clearance SDC-actor representation, do dataset-observable current-frame ODD-context proxies improve identification of ego-centric minimum swept OBB-TTC events, and are the resulting proxy associations coherent with temporal target profiles?

Our contributions are threefold:

1. Construct-Controlled Audit Formulation: We define a construct-controlled current-frame audit that compares a nearest-clearance focal baseline with nested ODD-context proxy families for an ego-centric all-nearby-actor TTC target; the target, proxy, and shared-state boundaries are explicit.
2. Sealed Holdout Incremental Finding: We apply a scenario-disjoint one-shot holdout protocol with scenario-equal metrics and paired scenario-block bootstrap inference, finding a supported but modest full-context increment ($\Delta\text{AP} = +0.0147$, 95% CI $[+0.0005, +0.0285]$). The protocol limits post-hoc tuning, preserves scenario disjointness, and makes the single holdout access auditable.
3. Feature-Family and Temporal Completeness Evidence: Family-restricted comparisons and development-to-holdout feature confirmation provide secondary localization evidence, while residualized scenario-profile analyses test temporal completeness against target Peak, AUC, and TET3. Warp experiments remain excluded and within-dataset vehicle-response KPIs are reported only as supportive development evidence.

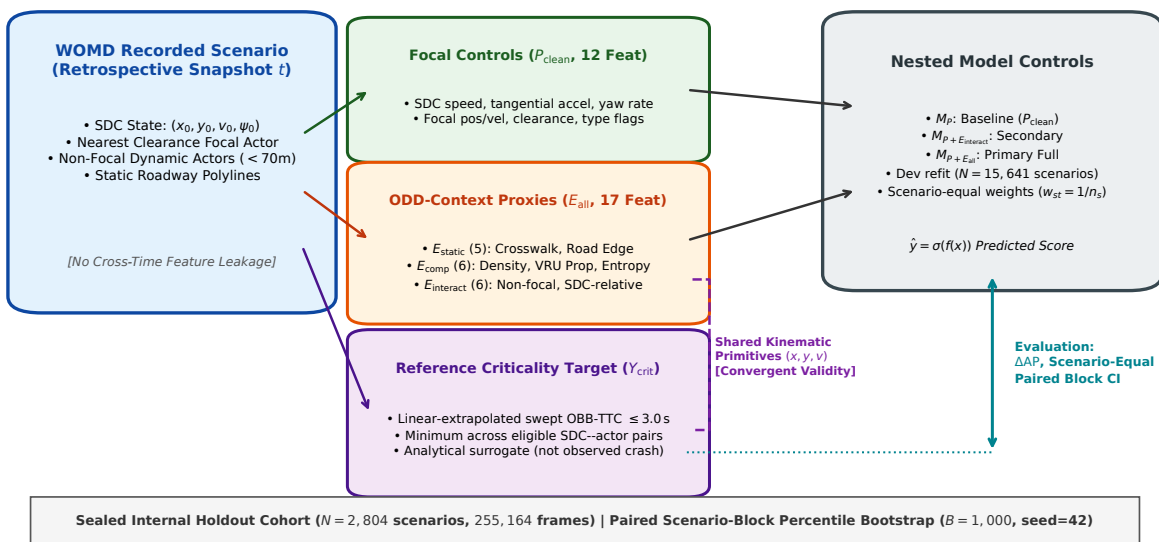


Figure 1. Evaluation Architecture and Construct Boundaries. Overview of the construct-controlled, scenario-disjoint audit protocol. Retrospective snapshot frames extract clean focal physical kinematics (P_{clean} , 12 features) and observable operational-context proxies (E_{all} , 17 features). Nested gradient-boosted models are evaluated against an ego-centric minimum swept SAT OBB-TTC reference target across eligible dynamic actors within 70 m on a sealed one-shot holdout cohort ($N = 255,164$ frames).

2. Related Work

2.1. Safety Surrogates in Autonomous Driving

Time-to-collision (TTC) and its variants (e.g., Modified TTC, Time-Exposed TTC) are foundational surrogates for traffic conflict severity [4, 7, 8, 15]. In automated vehicle development and large-scale public motion forecasting benchmarks (e.g., nuScenes [1], Argoverse [2], WOMD [3]), criticality metrics are widely used for mining scenarios and benchmarking models. SafeShift [12] explores safety-critical distribution shifts under geographical and environmental variations. Puphal et al. [9] propose risk-based filtering in WOMD. Weng et al. [14] emphasize joint multi-agent evaluation. However, benchmark evaluations frequently reduce scenes to focal agent pairs without quantifying what information is omitted.

2.2. ODD-Context Representation and Scene Analysis

Taxonomies for operational design domains (ODD) emphasize static geometry, dynamic actor mix, and environmental conditions [5, 6, 10, 13]. Query-centric and transformer-based motion forecasting architectures (e.g., MTR [11], QCNet [16]) aggregate multi-agent context via attention mechanisms. Together, these studies motivate interaction-aware evaluation;

our focus is complementary: we audit whether dataset-observable current-frame ODD-context proxies add predictive information beyond a focal SDC-actor kinematic baseline for an ego-centric TTC target.

3. Problem Formulation and Construct Boundary

3.1. Ego-Centric Target Estimand

Let $\mathcal{A}_{70}(s, t)$ denote eligible dynamic actors with valid current states within 70 m of the SDC in scenario s at frame t . The binary reference label is

$$Y_{s,t}^{(\tau)} = \mathbb{1}_{\left[\min_{j \in \mathcal{A}_{70}(s,t)} \text{TTC}_{\text{OBB}}(\text{SDC}, j) \leq \tau \right]}, \quad \tau = 3 \text{ s.} \quad (1)$$

Valid frames with $\mathcal{A}_{70}(s, t) = \emptyset$ are assigned $Y_{s,t}^{(\tau)} = 0$. The TTC is obtained by a constant-velocity swept OBB/SAT extrapolation from the state observed at frame t . Thus, “all-nearby-actor” refers to all eligible SDC-actor pairs, not arbitrary actor-actor pairs. The label is an analytical ego-centric surrogate, not an observed collision, causal risk, or system-safety ground truth.

For feature-level and temporal analyses, we addition-

119 ally use the continuous score

$$120 \quad C_{s,t} = \begin{cases} 1, & \text{TTC}_{s,t} = 0, \\ 1 - \text{TTC}_{s,t}/10, & 0 < \text{TTC}_{s,t} < 10 \text{ s}, \\ 0, & \text{otherwise,} \end{cases} \quad (2)$$

121 where $\text{TTC}_{s,t}$ is the minimum eligible SDC-actor swept
122 OBB-TTC; “otherwise” includes TTC of at least 10 s
123 and the absence of an eligible predicted contact. This
124 bounded score is an analysis surrogate and not a cali-
125 brated probability.

126 3.2. Shared-State Construct Boundary

127 The interaction proxies summarize non-focal actors rel-
128 ative to the SDC after excluding the nearest-clearance
129 focal actor. Both these proxies and the TTC reference
130 use current-state geometry and velocity. Their associ-
131 ation therefore represents shared-state convergent va-
132 lidity conditional on the focal controls; it must not be
133 interpreted as target-independent information, causal
134 influence, or closed-loop safety benefit.

135 3.3. Temporal Role: Retrospective Snapshot Anal- 136 ysis

137 Each WOMB training/validation sequence contains 91
138 samples over 9.0 s at 10 Hz: 10 history samples, one
139 current sample, and 80 future samples relative to the
140 benchmark prediction origin. Our analysis is retrospec-
141 tive. At each evaluated index t , predictors use only the
142 state recorded at that same index, and the TTC target
143 is a linear kinematic extrapolation from that state.

144 4. Data and Sealed Evaluation Protocol

145 4.1. Cohort Provenance and Sealed Split

146 We analyze all 18,445 WOMB v1.3.1 scenario parti-
147 tions discovered in the frozen local preprocessing in-
148 put; the evaluation pipeline applied no further sce-
149 nario sampling before splitting. The preprocessing arti-
150 facts do not retain upstream source-shard coverage as
151 probability-sampling metadata, so this cohort should
152 not be interpreted as representative of the full corpus.
153 A namespace- and seed-fixed SHA-256 assignment par-
154 titions the processed cohort by scenario ID into 12,828
155 training, 2,813 internal-validation, and 2,804 internal-
156 holdout scenarios; all 91 samples from a scenario re-
157 main in one split. Development comprises training plus
158 internal validation (15,641 scenarios). All model and
159 analysis choices were locked before the single holdout
160 inference pass.

161 4.2. Feature Taxonomy

162 P_{clean} contains 12 current-frame focal controls: SDC
163 speed, tangential acceleration, and yaw rate; focal rel-

164 ative longitudinal and lateral position and velocity; fo-
165 cal speed; OBB clearance and center distance; and
166 indicators for vehicle and vulnerable-road-user focal
167 types. The focal actor is selected by current OBB
168 clearance independently of the TTC outcome. The 17
169 ODD-context proxies comprise five static-roadway fea-
170 tures (distances to crosswalk, stop sign, speed bump,
171 and road edge, plus local lane-heading dispersion),
172 six traffic-composition features (actor counts within
173 30/50/70 m, vehicle and VRU proportions within 70
174 m, and actor-type entropy), and six non-focal SDC-
175 relative interaction features (nearest distance, mean
176 speed, speed standard deviation, heading dispersion,
177 and maximum/summed closing pressure).

4.3. Model Setup and Scenario-Equal Weighting

178 All nested models use HistGradientBoosting-
179 Classifier(max_iter=100, max_depth=6, ran-
180 dom_state=42) under scikit-learn 1.7.1; unspec-
181 ified parameters retain that version’s defaults.
182 Development-stage choices were made using the
183 training/internal-validation partition and then locked.
184 Each final frozen model was refit on all 15,641 devel-
185 opment scenarios using scenario-equal frame weights,
186 and the holdout pass contained zero fit calls. For
187 scenario s with n_s valid frames, each frame receives
188 $w_{s,t} = 1/n_s$, so every scenario contributes equal total
189 weight to AP, AUROC, and Brier score.
190

4.4. Statistical Inference Protocol

191 We use 1,000 paired scenario-block bootstrap repli-
192 cates with seed 42. Each replicate samples holdout
193 scenario IDs with replacement, retains every frame of
194 each sampled scenario, and evaluates both members
195 of a contrast on the identical draw. Within each oc-
196 currence, frame weights again sum to one per sampled
197 scenario. We report percentile 95% confidence inter-
198 vals for paired metric differences; we do not interpret
199 bootstrap sign-tail counts as calibrated p-values.
200

4.5. Individual Conditional Feature Effect Defini- 201 tion

202 For feature-level confirmation, we use continuous frame
203 severity $C_{s,t}$ rather than the binary threshold label. On
204 development data only, separate Ridge models with
205 $\alpha = 1$ regress each candidate proxy E_i and C on P_{clean} ,
206 using scenario-equal weights. Frozen nuisance models
207 produce holdout residuals $r(E_i)$ and $r(C)$; the reported
208 conditional effect is Spearman’s $\rho[r(E_i), r(C)]$. Confi-
209 dence intervals resample scenarios as blocks. The 13
210 candidates were frozen before holdout access.
211

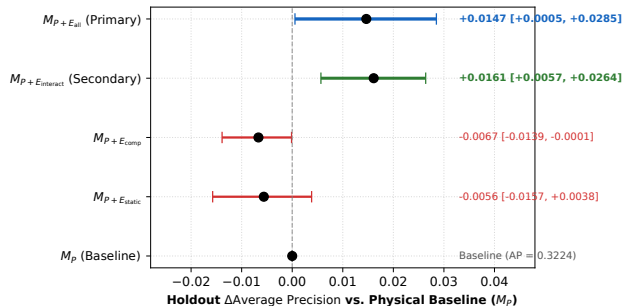


Figure 2. Forest Plot of Holdout Model Contrasts. Paired Δ AP and 95% scenario-block bootstrap confidence intervals relative to physical baseline M_P . The full context model $M_P + E_{\text{all}}$ (primary) and interaction model $M_P + E_{\text{interact}}$ (secondary) achieve strictly positive lower bounds.

5. Current-Frame Incremental Validity

5.1. Primary Holdout Evaluation

Table 1 presents the sealed holdout evaluation across nested model architectures on $N = 255, 164$ frames.

The primary contrast supports a positive but modest conditional increment: AP increases from 0.3224 to 0.3370 (Δ AP = +0.0147, 95% CI [+0.0005, +0.0285]). The lower bound is close to zero, so the result should not be described as a large or practically universal improvement. The context-only model is substantially weaker than the focal baseline (0.1141 vs. 0.3224), indicating that the proxies are useful conditionally rather than as a standalone replacement for focal kinematics.

5.2. Individual Feature Replication

Across the 13 pre-frozen candidate features evaluated in development, 10 out of 13 (76.9%) achieved strict holdout confirmation with 95% bootstrap confidence intervals strictly excluding zero in the expected direction. Furthermore, 13 out of 13 (100%) demonstrated exact sign concordance between development and holdout splits, as illustrated in Figure 3. Complete numerical breakdowns for all candidate features are provided in the Supplementary Material.

6. Feature-Family and Temporal Completeness Evidence

6.1. Interaction-Family Localization

Family-restricted comparisons localize the positive increment to the non-focal interaction family: $M_P + E_{\text{interact}}$ yields Δ AP = +0.0161 (95% CI [+0.0057, +0.0264]). Adding static features alone does not improve AP (Δ AP = -0.0056), while adding composition features yields a small decrease

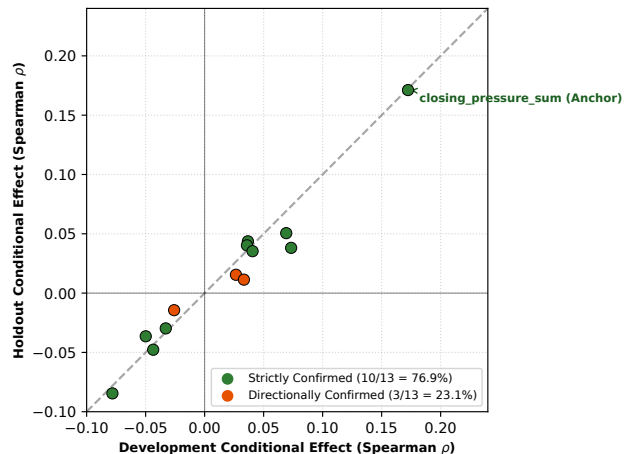


Figure 3. Feature Effect Concordance (Development vs. Holdout). Spearman rank correlation conditional effects across 13 candidate features showing 100% sign concordance and 76.9% strict confirmation.

(Δ AP = -0.0067). These nested comparisons are consistent with interaction summaries accounting for the observed increment within the evaluated model family; they do not identify a causal mechanism or establish that map and composition context are generally irrelevant.

6.2. Scenario Profile Temporal Completeness

We use scenario profiles as a temporal completeness check, not as the primary estimand. For each proxy i , we compute its scenario mean $\bar{E}_{s,i} = T_s^{-1} \sum_t E_{s,t,i}$. From continuous frame severity $C_{s,t}$ and TTC, we form three target profiles: $D_s^{\text{Peak}} = \max_t C_{s,t}$, $D_s^{\text{AUC}} = \sum_t C_{s,t} \Delta t$, and $D_s^{\text{TET3}} = \sum_t \mathbb{I}[\text{TTC}_{s,t} \leq 3 \text{ s}] \Delta t$. Separate Ridge models with $\alpha = 1$, fitted on development scenario means of P_{clean} , residualize $\bar{E}_{s,i}$ and each target profile. We report holdout Spearman correlations between the corresponding residuals with scenario-level bootstrap intervals.

The pre-specified closing-pressure-sum anchor is positive at the frame level ($\rho = +0.1712$) and for target Peak (+0.1914, 95% CI [+0.1549, +0.2300]), target AUC (+0.2345, [+0.2040, +0.2677]), and target TET3 (+0.0728, [+0.0405, +0.1086]). These results support anchor-level temporal completeness, but the heterogeneous magnitudes—particularly the smaller TET3 association—and the mixed 17-feature taxonomy warrant a feature-dependent, not uniformly strong, conclusion. Across the full 17-feature taxonomy, 8 features are classified as CROSS_LEVEL_STABLE, 6 as UNSUPPORTED_BOTH, 1 as FRAME_LOCAL, 1 as DISCORDANT_SIGN, and 1 as TEMPO-

Table 1. Primary Nested Model Evaluation on Sealed Holdout Cohort ($N = 255,164$ frames, 2,804 scenarios). Models trained on all 15,641 development scenarios and evaluated on the sealed holdout. Primary contrast is $M_{P+E_{all}}$ vs. M_P . Confidence intervals are pre-specified 95% paired scenario-block percentile bootstrap intervals ($B = 1,000$).

Model	Feature Set	Dim	AP	AUROC	Brier	Δ AP vs. M_P	95% CI
M_P (Physical Baseline)	P_{clean}	12	0.3224	0.8022	0.0451	Baseline	—
M_E (Context Only)	E_{all}	17	0.1141	0.7113	0.0516	-0.2082	[-0.2241, -0.1925]
$M_{P+E_{static}}$	$P_{clean} + E_{static}$	17	0.3167	0.8046	0.0452	-0.0056	[-0.0157, +0.0038]
$M_{P+E_{comp}}$	$P_{clean} + E_{comp}$	18	0.3157	0.8078	0.0453	-0.0067	[-0.0139, -0.0001]
$M_{P+E_{interact}}$ (Secondary Core)	$P_{clean} + E_{interact}$	18	0.3385	0.8347	0.0444	+0.0161	[+0.0057, +0.0264]
$M_{P+E_{all}}$ (Primary Full)	$P_{clean} + E_{all}$	29	0.3370	0.8399	0.0444	+0.0147	[+0.0005, +0.0285]

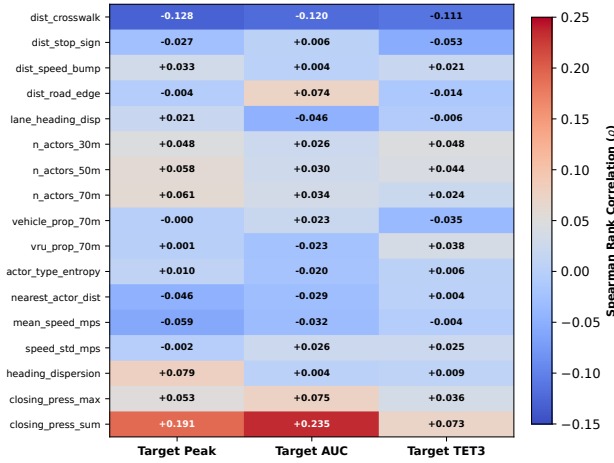


Figure 4. Temporal Target-Profile Completeness. Holdout residual Spearman associations between each scenario-mean ODD-context proxy and target Peak, AUC, and TET3 after development-fitted adjustment for scenario-mean focal kinematics ($N = 2,804$ scenarios). Column labels refer to target profiles, not transformations of the proxy itself.

RAL_EXPOSURE_SPECIFIC.

7. Discussion, Limitations, and Intended Use

7.1. Threshold Horizon Sensitivity

Sensitivity results establish a horizon boundary rather than a behavioral cause. The context increment is positive at 3 s (Δ AP = +0.0147), larger at 5 s (Δ AP = +0.0660), and negative in the rarer 2 s tail (Δ AP = -0.0442). The available analysis does not determine whether this variation arises from prevalence, calibration, model capacity, or a change in the underlying event regime; we therefore avoid attributing the 2 s result to post-encroachment reaction behavior.

7.2. Supportive Development KPI Construct Validity

Within-dataset vehicle-response KPIs provide supportive development-only convergent evidence. For exam-

ple, the continuous TTC-based severity score $C_{s,t}$ is modestly associated with SDC hard-deceleration p95 (Spearman $\rho = +0.1347$; Cohen's $d = +0.3451$). Other KPIs are weaker or directionally mixed, including the DRAC summaries. Because these KPIs are computed from the same recorded sequences and share kinematic state, they are neither external validation nor evidence of crash avoidance.

7.3. Limitations and Excluded Explorations

This study is a retrospective audit of analytical surrogates on a processed WOMB cohort. The TTC target assumes constant current velocity and OBB geometry and is not an observed crash label. Predictors and target share current-state primitives, so the evidence is convergent rather than orthogonal or causal. The processed 18,445-scenario cohort is not the complete WOMB corpus, and the internal holdout does not establish cross-dataset, geographic, or population validity. Recorded trajectories cannot evaluate closed-loop policy response or the safety performance of any deployed driver. Threshold sensitivity shows that the primary conclusion does not extend unchanged to every TTC horizon. Warp experiments were inconclusive and are excluded; within-dataset KPI analyses remain supportive and development-only.

8. Conclusion

We presented a scenario-disjoint audit of whether current-frame ODD-context proxies add conditional validity beyond a nearest-clearance focal-kinematic baseline for an ego-centric all-nearby-actor TTC surrogate. On a one-shot holdout, the full proxy set produced a supported but modest AP increment (Δ AP = +0.0147, 95% CI [+0.0005, +0.0285]). Family-restricted comparisons localize the increment to non-focal SDC-relative interaction summaries, whereas static and composition families do not improve AP when added alone. The pre-specified interaction anchor also remains positive across residualized target Peak, AUC, and TET3 profiles, with feature-dependent support across the full

taxonomy. Within the processed cohort, these findings provide a bounded evaluation result: focal scene reductions can omit measurable current-frame context. They do not establish population-wide WOMD validity, causal risk, crash prediction, or closed-loop system safety.

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